Evaluating the success of bus franchising in Greater Manchester

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1. Introduction

In his journal article about English devolution, Cox frames the topic in terms of 'the English Question', namely the idea that England is overcentralised (Cox, 2016). Civic nationalist movements in Scotland, Wales and Northern Ireland have granted other countries in the Kingdom significant decision-making powers via the Scottish Parliament, National Assembly for Wales and the Northern Ireland Assembly respectively, each of which is somewhat independent of Whitehall. There is a clear desire for similar devolved powers to be granted to regions of England itself, with Manchester having a strong sense of pride and cultural identity (Coleman, Segar and Checkland, 2015) and Yorkshire having had a distinct regionalist movement for over a decade (Giovannini, 2016).

Cox makes clear the guardedness of Whitehall in granting devolutionary powers to English regions, noting that "when cities have attempted to broker devolution of key aspects of local economic development such as education, key areas of skills and innovation, housing, welfare and employment support, government departments have drawn red lines and denied significant local autonomy." (Cox, 2016, p. 567). In their book evaluating the success of metro-mayors in Greater Manchester and the Liverpool City Region, Blakeley and Evans state that policymaking power still ultimately lay with Whitehall, even arguing that the combined authorities and metro-mayors have less power than the city councils have in their respective regions (Blakeley and Evans, 2023, ch. 10).

However, we shouldn't overly minimise the role of combined authorities in allowing for a higher level of place-based policymaking and potentially contributing to welfare and growth in their regions. It may not wholly be a coincidence that the decade in which the GMCA has been granted greater autonomy coincides with Manchester outstripping the growth rate of the UK as a whole (Spencer, 2025). Moreover, when considering the future course of combined authorities, Blakely and Evans argue that, based on "empirical research conducted into the activities of the GCMA and LCRCA", it is most likely that CAs will enjoy an "incremental growth of … power and resources" (Blakeley and Evans, 2023, pp. 228–229). As such, I believe it is important to investigate the efficacy of the policies put in place in the GMCA. In doing this, I hope to move closer to an answer for the question of whether place-based policymaking is a positive direction for the English political system to move towards.

2. Theory

We will analyse the franchising of buses in the GMCA. Between 2023 and 2025, a full franchising model for Greater Manchester's bus network was rolled out, with the process completed in January 2025. While potentially not a focus for many economists, transport is a huge aspect of local policymaking. Bakely and Evans note that it has a close connection to inclusivity through multiple channels, namely helping to tackle social isolation by providing more mobility opportunities to the elderly and disabled, allowing adults and young people to secure training and employment as well as cleaner forms of transport potentially diminishing pollution for poorer residents in city centres. Burnham himself, the metro-mayor of Greater Manchester, asserts that re-regulation of buses specifically would improve equality, open up skills training possibilities to young workers, reduce air pollution and generally enhance economic development (Blakeley and Evans, 2023, pp. 74, 84).

In their paper analysing approaches to improving bus services, Rye et al. define bus franchising as "re-regulation of the services in an area under public sector control, with operator(s) running services under contract in a model similar to that in Scandinavia or London" (Rye *et al.*, 2021, p. 60). Critics of fully privatised bus network argue that bus companies in Greater Manchester were using local monopoly status to profiteer, rather than providing the bus services that the community needed. For example, Jim McMahon, MP for Oldham West and Royton, accused the two dominant bus companies, namely Stagecoach and First Bus, of being private sector monopolies which ran 90 per cent of passenger journeys to their own benefit. Moreover, Debbie Abrams, MP for Oldham East and Saddleworth, alleged that the privately run bus companies had delivered 25 million fewer journeys in five years (Blakeley and Evans, 2023, pp. 82–83). If one views public transport as a merit good (or if it facilitates people consuming other merit goods such as education and healthcare), meaning that its benefits to society are higher than that which can be captured monetarily by the private sector, then it can be argued that private bus companies will under-provision bus services if left under-regulated. Introducing stricter regulation and heavier public-sector involvement may remedy this.

In a 2018 paper, Cowie counters this idea. His empirical study determines that bus companies outside of London have been making close to normal profits since the global financial crisis, meaning that it is unlikely that there is slack, or x-inefficiency, and that re-regulation is likely to lead to similar results to that which exist with deregulated bus operations (Cowie, 2018). A later study with a wider remit came to a similar conclusion, concluding that the economics of bus operation has tightened since privatisation due to economy-wide factors; bus re-regulation is unlikely to fix the root causes of decreasing bus patronage, at least not without other central government measures (Xavier *et al.*, 2022).

I am going to explore whether the place-based policy of bus franchising in Greater Manchester has led to improved outcomes for citizens of the combined authority, indicative of a successful policy. The aim of this is to use the outcome of this study as a yardstick to measure how well they are able to implement policy in general. My hypothesis is that greater devolution of transport powers allows for successful place-based policymaking, improving transport outcomes for locals. Specifically, devolution has allowed the GMCA to improve the bus network in Greater Manchester.

3. Data and Methods

My first data source is Local Authority-level bus journey panel data from the Department for Transport (Department for Transport, 2025b, tab: BUS01f). The dataset contains data on the per-capita number of passenger bus journeys taken in each local authority in each year. My dependent variable is the number of passenger journeys on local bus services per head for each local authority. My main independent variable is a policy indicator which is equal to one if buses are franchised (or partly franchised) in that local authority in that year, and zero otherwise. Over the years that these data are collected, buses have only been gradually franchised in Greater Manchester over the years 2023-2025 (Centre for Cities, 2024), so the policy indicator is only equal to one for two of the data points.

In each part of the following analysis, I exclude the Greater London Authority as I believe it is an outlier which belongs in a different category to the rest of England. One reason for this is that the bus sector in London wasn't ever privatised in the same way as it was in the rest of England, meaning that it is difficult to treat London in the same way as the rest of the country in this context. Considering 'England outside of London' is standard practice in the relevant literature.

The suitability of each of these models in producing unbiased and consistent estimators is evaluated in Section 3 of the Appendix.

For the first part of this analysis, I use the general framework for policy analysis with pooled cross sections set out in Wooldridge's introductory econometrics textbook (Wooldridge, 2025, pp. 455–457). Here, the estimate of the parameter multiplying the policy indicator, labelled $treatment_{i,t}$ below, is the estimated ceteris paribus effect of the policy. By allowing the intercept of the equation to change across time periods, this framework includes aggregate time effects which control for external factors that affect all Local Authorities equally.

$$\text{bus\_journeys\_per\_capita}_{i,t} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta \,\text{treatment}_{i,t} + u_{i,t}$$
$$\text{for } i = 1, \ldots, 89, t = 1, \ldots, 16 \text{ (where applicable)} \tag{1}$$

Figure 3.1 | Functional form for the initial pooled OLS regression of the number of bus journeys taken per person per year on the treatment indicator (which is equal to 1 if buses are franchised in that local authority at that time, and 0 otherwise), as well as time dummy variables for each time period [model (1)]

To mitigate omitted variable bias, we need to take factors out of the error term that the treatment indicator may be correlated with. As such, in the second part of this analysis I control for some factors that affect the demand for bus services and may be correlated with whether buses are franchised in a given region. Among the main factors that affect demand for bus services are income, car ownership, pricing of bus service, pricing of competing and complementary public transport services, quality of the bus service, demographic makeup of the area and population density of the area (Balcombe *et al.*, 2004, ch. 3). I do not control for the pricing nor quality of the bus service. These variables are 'bad controls' as we are trying to measure how bus franchising has affected people's willingness to use the service; by including pricing and quality of service in the regression, I would be controlling for exactly what we are trying to measure. I also do not control for demographic makeup of the regions nor for population density of the regions. This is because these data are not yet available sufficiently up-to-date from the ONS (Office for National Statistics, 2025a).

So, I aim to control for income, car ownership levels and pricing of competing and complementary public transport services. I use as a measure of income the median annual earnings for each region, published yearly by the ONS (Office for National Statistics, 2025b, tbl. 8.7a, tab: All). These nominal wage data were adjusted to real wage data using inflation statistics from the ONS (Office for National Statistics, 2026). Panel data were not available on the number of cars owned per household in each local authority, but as a proxy variable I use the total number of private (non-company owned) licensed cars in each local authority in each year, published by the Department for Transport (Department for Transport, 2026). The main form of public transport operating in Greater Manchester, aside from ride-hailing apps, is the Manchester Metrolink tram/light rail network. While data on pricing on light rail and tram networks across England wasn't available, I use as a proxy variable the passenger journeys each light rail and tram network in England, also published by the Department of Transport (Department for Transport, 2025a, tbl. lrt0101).

$$\text{bus\_journeys\_per\_capita}_{i,t} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta\, \text{treatment}_{i,t} + \gamma_1\, \text{real\_med\_income\_pc}_{i,t} + \gamma_2\, \text{priv\_lic\_cars}_{i,t} + \gamma_3\, \text{tram\_jours\_percap}_{i,t} + u_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable) (2)

Figure 3.2 | Functional form for the pooled OLS regression of the number of bus journeys taken per capita per year on the treatment indicator, the real median per-capita income of each local authority, the total number of private licensed cars in each local authority (thousands) and the number of tram journeys per person per year in each local authority [model (2)]

To further reduce omitted variable bias in the treatment indicator and pin down the ceteris paribus effect of the bus franchising policy on the use of the buses, I use the general policy analysis with panel data framework set out in the Wooldridge textbook (Wooldridge, 2025, pp. 497–498), estimating the following equation via 'two-way fixed effects'.

Unobserved effects equation:

$$\text{bus\_journeys\_per\_capita}_{i,t} = \alpha + \beta\, \text{treatment}_{i,t} + \gamma_1\, \text{real\_med\_income\_pc}_{i,t} + \gamma_2\, \text{priv\_lic\_cars}_{i,t} + \gamma_3\, \text{tram\_jours\_percap}_{i,t} + a_i + u_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Two-way fixed effects transformation:

$$\text{bus\_journeys\_per\_capita}_{i,t} - \overline{\text{bus\_journeys\_per\_capita}_i} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta(\text{treatment}_{i,t} - \overline{\text{treatment}_i}) + \gamma_1(\text{real\_med\_income\_pc}_{i,t} - \overline{\text{real\_med\_income\_pc}_i}) + \gamma_2(\text{priv\_lic\_cars}_{i,t} - \overline{\text{priv\_lic\_cars}_i}) + \gamma_3(\text{tram\_jours\_percap}_{i,t} - \overline{\text{tram\_jours\_percap}_i}) + (u_{i,t} - \overline{u_i})$$

where $\overline{k_i}$ denotes the time average of k, $\overline{k_i} = \frac{1}{T}\sum_{t=1}^{T} k_{i,t}$, or more simply

$$\ddot{\text{bus\_journeys\_per\_capita}}_{i,t} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta\, \ddot{\text{treatment}}_{i,t} + \gamma_1\, \ddot{\text{real\_med\_income\_pc}}_{i,t} + \gamma_2\, \ddot{\text{priv\_lic\_cars}}_{i,t} + \gamma_3\, \ddot{\text{tram\_jours\_percap}}_{i,t} + \ddot{u}_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable) (3)

Figure 3.3 | The same regression functional form as in Figure 3.2, but with the two-way fixed effects transformation applied [model (3)]

Now, we are explicitly allowing the franchising of buses to be correlated with variables that are constant over time (a_i in the above unobserved effects model). By time demeaning each of the variables in the equation, the two-way fixed effects transformation removed these time-constant variables from the model. In addition to this, the inclusion of time-period dummies in the regression, just as with the previous pooled OLS regression, allows for other aggregate time factors to affect the dependent variable; by using the two-way fixed effects transformation we are controlling for systematic differences in the cross sectional units that do not vary across time, as well as systematic differences across time that do not vary by unit.

As a robustness check, I perform a first differencing regression on the data. Similar to the fixed effects transformation, the first differencing transformation removes the fixed effects from the equation, but it does so by differencing the variables across time rather than by time demeaning the variables. If the first differencing regression's results agree with the fixed effect regression's results, this adds robustness to the result of the regression.

Unobserved effects equation:

$$\text{bus\_journeys\_per\_capita}_{i,t} = \alpha + \beta\, \text{treatment}_{i,t} + \gamma_1\, \text{real\_med\_income\_pc}_{i,t} + \gamma_2\, \text{priv\_lic\_cars}_{i,t} + \gamma_3\, \text{tram\_jours\_percap}_{i,t} + a_i + u_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

First differencing transformation (with aggregate time effects):

$$\Delta\text{bus\_journeys\_per\_capita}_{i,t} = \delta_2 \Delta\text{d2011}_t + \delta_3 \Delta\text{d2012}_t + \ldots + \delta_{16} \Delta\text{d2025}_t + \beta\, \Delta\text{treatment}_{i,t} + \gamma_1\, \Delta\text{real\_med\_income\_pc}_{i,t} + \gamma_2\, \Delta\text{priv\_lic\_cars}_{i,t} + \gamma_3\, \Delta\text{tram\_jours\_percap}_{i,t} + a_i + \Delta u_{i,t}$$

for i = 1, ..., 89, t = 2, ..., 16 (where applicable) (4)

Figure 3.4 | The same regression functional form as in Figure 3.2, but with the first differencing transformation applied [model (4)]

I now try to reduce the potential over-parameterisation in my model. Due to having a relatively small sample size and large number of regressors (N = 89, k = 19 for the fixed effects regression), removing the aggregate time effects dummies, while still trying to partial out systematic factors that vary across time but not by cross sectional unit, might lend more credibility to the model. As such, I remove the time period dummies and add in a linear time trend t, as well as adding in a dummy variable, *covid*, which is equal to 1 for all units where the year is equal to 2021 or 2022 (representing year ending March 2021 and March 2022 respectively). As we can see by examining the value of the time dummy variables in the regressions (see Table 4.1), these years represent huge drops in bus patronage due to coronavirus restrictions and people subsequently being less likely to use public transport.

Unobserved effects equation:

$$\text{bus\_journeys\_per\_capita}_{i,t} = \alpha + \beta\,\text{treatment}_{i,t} + \gamma_1\,\text{real\_med\_income\_pc}_{i,t} + \gamma_2\,\text{priv\_lic\_cars}_{i,t} + \gamma_3\,\text{tram\_jours\_percap}_{i,t} + \psi_1 t + \psi_2\,\text{covid}_t + a_i + u_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Fixed effects transformation:

$$\ddot{\text{bus\_journeys\_per\_capita}}_{i,t} = \beta\,\ddot{\text{treatment}}_{i,t} + \gamma_1\,\ddot{\text{real\_med\_income\_pc}}_{i,t} + \gamma_2\,\ddot{\text{priv\_lic\_cars}}_{i,t} + \gamma_3\,\ddot{\text{tram\_jours\_percap}}_{i,t} + \psi_1 \ddot{t} + \psi_2\,\ddot{\text{covid}}_t + \ddot{u}_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

(5)

Figure 3.5 | The same regression functional form as in Figure 3.3, but with the time dummy variables omitted and replaced with a linear time trend and a pandemic dummy variable, equal to 1 in the years (ending March) 2021 and 2022 and equal to zero otherwise [model (5)]

This model is less flexible than the two-way fixed effects model, but importantly I am conserving degrees of freedom: the model has only six independent variables, whereas the two-way fixed effects model has nineteen.

In my analysis, I use heteroscedasticity- and cluster-robust standard errors, as is standard practice when working with panel data.

4. Results

4. a. Initial Finding

Taking an initial glance at the numbers, it seems that the policy has a positive effect on the number of bus journeys taken in the GMCA.

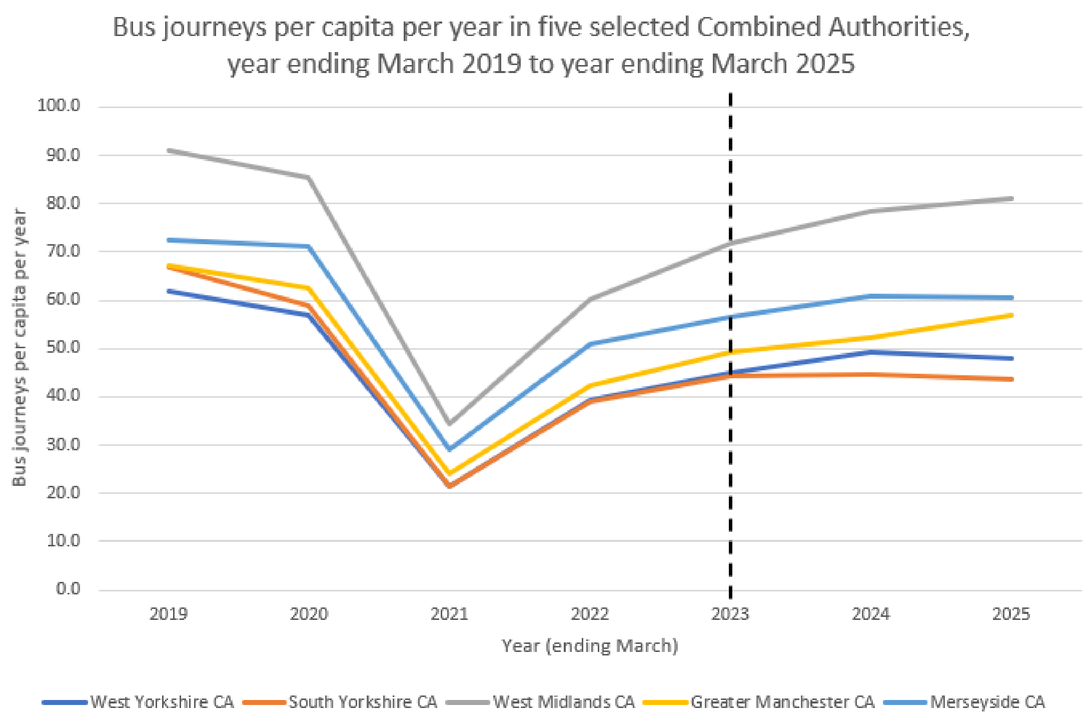

Figure 4.1 | A graph of the number of per-person per-year bus journey taken in five Northern/Midlands Combined Authorities, with a vertical line separating the pre- and post-treatment period for the GMCA. Data are taken from the Department for Transport's 'Local bus passenger journeys (BUS01)' dataset (Department for Transport, 2025b).

When comparing the number of bus journeys taken in Greater Manchester alongside comparable Northern and Midlands Combined Authorities, by March 2025 bus journeys seem to be increasing faster than in South Yorkshire, West Yorkshire and Merseyside, while increasing at a comparable pace to the West Midlands. No conclusions can be drawn here, so we now move to using a formal policy analysis framework.

4. b. Multiple Linear Regression results

Model:	(1) Pooled OLS with no controls	(2) Pooled OLS with controls	(3) Two-way fixed effects with controls	(4) First differencing with controls	(5) One-way fixed effects with linear time trend
Regressors					
$treatment_{i,t}$	19.689***	-55.510**	-14.410*	-5.099*	-16.379**
	(2.818)	(21.693)	(7.351)	(3.013)	(7.730)
$real\_med\_income\_pc_{i,t}$	-	-0.002**	-0.0002	-0.0003	-0.0002
		(0.0009)	(0.0005)	(0.0004)	(0.0004)
$priv\_lic\_cars_{i,t}$	-	-0.014	-0.067*	-0.144***	-0.062
		(0.014)	(0.037)	(0.047)	(0.038)
$tram\_jours\_percap_{i,t}$	-	1.988***	0.927**	1.127**	1.072**
		(0.536)	(0.443)	(0.497)	(0.458)
t	-	-	-	-	-0.840***
					(0.146)
$covid_t$	-	-	-	-	-16.681***
					(1.327)
R^2	0.0975	0.2727	0.0561	0.6321	0.5983
Total no. observations	1,368	1,359	1,359	1,272	1,359

*Table 4.1 | Regression results for each of the five models from Section 3, reported alongside heteroscedasticity- and cluster-robust standard errors in brackets below. Coefficients are significant at the *10%, **5% and ***1% levels.*
A constant, as well as dummy variables for each time period, are included in each model (time dummy variables omitted in model (5)) but are not reported.

The models (3) to (5) likely produce the estimates with the least bias, as discussed in Section 3. The result from the two-way fixed effect regression including control variables (model (3)), suggests that bus franchising has a negative ceteris paribus effect on the number of bus journeys taken per head, with bus franchising reducing the number of journeys taken by ~14.4 per person per year.

The results from the first differencing model largely agree with that of the fixed effects regression: the results are not sensitive to the method of removing the region-specific fixed effects. The main difference in the result is that the estimate of the treatment effect is smaller, suggesting that people take ~5.1 less journeys per year due to bus franchising rather than ~14.4 less.

In the final degrees of freedom-conserving model (model (5)), the main results of the two-way fixed effects regression are maintained: there is a negative, statistically significant impact of the treatment indicator on the number of bus journeys taken per year per person.

5. Limitations

This study has numerous limitations which mean one should exercise caution in accepting a causal interpretation of the effect of the bus franchising policy on the number of bus journeys taken in Greater Manchester.

The most obvious and largest problem with this study is the lack of post-treatment data. As the policy was only rolled out between 2023 and 2025, and data is only available up to March 2025, the policy indicator is only equal to one for 2 out of the 1,359 datapoints in the two-way fixed effects estimated model. This treatment sample size is far too small to come to any conclusion on the efficacy of the policy, which threatens the internal validity of the study. Related to this, as the policy was brought in gradually between 2023 and 2025, assigning a binary treatment effect may not strictly be the most accurate method to parse out a treatment effect for this policy.

Related to the above, the robustness of the result would be strengthened greatly if there were more treatment units. GMCA is the only local authority to franchise their bus network thus far; internal validity (with respect to England) of the result would be improved greatly if other local authorities had also taken on this policy.

There are further issues with this study, namely concerns over omitted variable bias, proxy variable bias, missing data bias and doubts over asymptotic justification of inference. These are taken up in section 5 of the Appendix.

6. Conclusion

Taking the results of these linear regression models at face value, one can only conclude that bus franchising has had a modest negative effect, or no effect, on the number of passenger bus journeys taken in Greater Manchester.

The regression policy analysis framework suggests that the bus franchising policy has had a modest negative effect on the number of bus journeys taken, reducing it by between 5 and 16 bus journeys per capita per year. However, this conclusion isn't strong as standard statistical significance and robustness criteria aren't met for these models.

Despite the lack of a clear conclusion, I believe the policy analysis framework set out here is strong and that this study would lead to a more convincing conclusion if conducted in five or ten years' time. This is because there will be further years of data available on Greater Manchester's bus journey numbers, as well as data on other local authorities who are planning to roll out franchising policies over the next few years such as West Yorkshire, South Yorkshire, the West Midlands and the Liverpool City Region (Centre for Cities, 2024). Moreover, there will be more up-to-date ONS data on control variables such as demographic information and population density of each local authority. With more data available, a clearer picture is likely to emerge regarding the efficacy of bus franchising in England. If data could be collected on the local authorities making up each combined authority, there would be more cross-sectional units making up the sample, lending more credibility to asymptotic justifications of test statistics, as well as there being a larger treatment sample.

There are some final considerations to be made regarding drawing conclusions from this study. We have here looked at the narrow, empirical question of whether the franchising of buses has increased the number of journeys taken in Greater Manchester. While this is an indication of the success of devolution and place-based policymaking, one may reasonably argue that we are asking the wrong question. It could be argued that the primary objective of the policy analysed here, and what most voters care about, is that buses are now more tightly controlled by the public sector, rather than any secondary objectives such as an immediately improved experience or a greater number of bus routes. Viewed through this lens, the bus franchising policy has clearly been a success.

Even more than this, it can be argued that devolution and strong local leadership were key to buses being devolved in Greater Manchester. In 2011, prior to 2017's Bus Services Act which allowed for mayoral combined authorities to franchise buses without the previous requirement to show market failure (Xavier *et al.*, 2022, p. 2), the North East Combined Authority (NECA) put forward a proposal to re-franchise the regions bus network. This proposal ultimately did not go ahead, with regulators deciding that the scheme would have been too costly for councils to afford and would not have increased the use of bus services; interviewees from NECA noted that the bus companies had much larger and more well-funded legal teams, as well as better access to market information, putting them at a disadvantage when making their case (Rye *et al.*, 2021, sect. 5). Meanwhile, in the GMCA Burnham had to fight and make tough choices to franchise the buses, including raising the mayoral precept to fight a legal challenge from the bus companies operating in the GMCA (Blakeley and Evans, 2023, ch. 5). With fewer devolved powers and a less determined metro-mayor, GMCA's bus franchising may never have been allowed.

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A. Appendix

Section 1: Introduction

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Section 2: Theory

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Section 3: Data and Methods

All data used in the Data and Methods section of this paper is collected from reputable UK Government sources, namely the Office for National Statistics and the Department for Transport. As such, there are few data reliability concerns for this study.

Now, I consider how well my regression models match up with the assumptions necessary for them to produce consistent estimates of the treatment effect, as well as the assumptions necessary for valid test statistics to be constructed.

I first note that it is unlikely that either model (1) nor model (2) produce consistent estimates of the ceteris paribus effect of the franchising of buses on the number of passenger bus journeys taken in Greater Manchester. The main assumption needed for the regression in model (1) is that the treatment indicator is uncorrelated with the error term, $u_{i,t}$ in every time period (Wooldridge, 2025, ch. 5.1, 10.3). The error term contains factors that affect the number of bus journeys taken by people in the local authorities that aren't included among the regressors in the above equation – these could be the local authority's level of car ownership, availability of other forms of public transport, income levels, and even the population's political leanings. Even just considering contemporaneous exogeneity this is a strong assumption, and in fact it is more realistic to believe that bus franchising is to be correlated with factors such as the above. This introduces a potentially (very likely) large bias into the result from this pooled OLS regression.

For model (2) to produce a consistent estimate of the capture the ceteris paribus effect of the franchising of buses on the number of passenger bus journeys taken in Greater Manchester, the main assumption is still that the treatment indicator is uncorrelated with the error term, $u_{i,t}$ in every time period. Now, income, private car ownership and journeys taken on competing or complementary transport systems have been taken out of the error term, so this is a slightly weaker assumption. But the error term still contains many city-specific variables that may be correlated with the demand for bus services as well as whether buses are franchised: these may include path-dependency, or the tendency for people to take certain types of transport in a region just because they are used to that form of transport, political leanings of the population, demographic makeup of the region and population density of the region. As such, it is likely that there is still bias in the estimate of the treatment effect, as well as in the estimate of the effects of the control variables.

I now evaluate model (3) against each of the assumptions needed for fixed effects estimators to be unbiased and consistent: these assumptions are laid out in the Wooldridge textbook (Wooldridge, 2025, pp. 512–514). The first assumption (FE.1) is that the model is linear parameters (i.e. it takes the form $y_{i,t} = \beta_1 x_{i,t} + \ldots + \beta_k x_{ik} + a_i + u_{i,t}$, $i = 1,\ldots,n$, $t = 1,\ldots,T$). While I haven't explored the possibility of nonlinear effects of the regressors on the dependent variable in this paper (limitations of degrees of freedom due to a small cross-sectional sample size arguably preclude this possibility), I think it is a weak assumption to make that average real incomes, the number of licenced cars in a local authority and the number of tram journeys taken per person have a linear effect on the number of bus journeys taken per capita over a standard range of values.

The second assumption (FE.2) is that we have a random sample from the cross section. This assumption is clearly violated, as we are using aggregated, local authority-level data rather than sampling small units randomly from a large population.

The third assumption (FE.3) is that each regressor changes over time for at least some i, and there is no perfect collinearity among the regressors. This is a very weak assumption and is true in the case of this model.

The final and most important assumption strict exogeneity (FE.4): for each t, the expected value of the idiosyncratic error ($u_{i,t}$) given the regressors in all time periods and the unobserved effect is zero: $E(u_{i,t} \mid \underline{X}_i, a_i) = E(u_{i,t}) = 0$. For this to be true, time-varying variables in the error term must be uncorrelated with past, present and future values of the explanatory variables (and the time constant omitted variables; this is true by construction). This assumption is likely violated – even contemporaneous exogeneity may be violated if the treatment indicator or the regressors are correlated with factors such as population density or demographic factors. However, the fixed effects transformation eliminates factors such as path dependency or political leanings from the error term as they are likely to be approximately fixed over time. This means that this assumption is far likelier to be approximately true than in the case of model (1) or (2).

We do not need to make assumptions FE.5 (heteroscedasticity) nor FE.6 (no autocorrelation in idiosyncratic errors) as I use standard errors and test statistics that are heteroscedasticity- and cluster-robust. I do not make assumption FE.7 (normally distributed errors) as I rely on asymptotic approximations – these approximations rely on a large cross sectional and small time series sample size. With N = 89 and T = 16 in my sample, this is likely to be overly optimistic.

The assumptions necessary for models (4) and (5) to be unbiased and consistent are very similar to the ones discussed above. Please see Section 5 of the main text and of the Appendix for more considerations regarding the unbiasedness and consistency of the estimators in this paper.

Section 4: Results

Please see the GitHub repository for the full code for this project. I have printed screenshots from the relevant output of the code below.

Functional Form for simple Pooled OLS:

$$\text{bus\_journeys\_per\_capita}_{i,t} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta\, \text{treatment}_{i,t} + u_{i,t}$$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Results (with heteroscedasticity and cluster-robust SEs):

```
> summary(a) ##This prints the regression results
Pooling Model

Call:
plm(formula = Bus.Journeys.Per.Capita.Per.Year ~ Treatment +
    a.1 + a.2 + a.3 + a.4 + a.5 + a.6 + a.7 + a.8 + a.9 + a.10 +
    a.11 + a.12 + a.13 + a.14 + a.15, data = BusJourneys_PrivateCars_TramJourneys_RealPay,
    model = "pooling", index = c("LA.or.Region", "Year"))

Unbalanced Panel: n = 87, T = 5-16, N = 1368

Residuals:
   Min. 1st Qu.  Median 3rd Qu.    Max.
 -43.45  -18.12   -8.07   11.29  135.54

Coefficients:
             Estimate Std. Error t-value  Pr(>|t|)
(Intercept)  49.08410    3.01584 16.2754 < 2.2e-16 ***
Treatment    19.68941   19.65818  1.0016 0.3167218
a.1          -0.80603    4.26504 -0.1890 0.8501326
a.2          -2.13215    4.25248 -0.5014 0.6161782
a.3          -3.89570    4.25248 -0.9161 0.3597775
a.4          -3.48976    4.25248 -0.8206 0.4119949
a.5          -4.06689    4.25248 -0.9564 0.3390626
a.6          -4.72398    4.25248 -1.1109 0.2668194
a.7          -5.27461    4.25248 -1.2404 0.2150574
a.8          -7.00221    4.25248 -1.6466 0.0998689 .
a.9          -7.22275    4.25248 -1.6985 0.0896476 .
a.10         -9.87291    4.25248 -2.3217 0.0203981 *
a.11        -35.05352    4.22811 -8.2906 2.691e-16 ***
a.12        -22.09806    4.22811 -5.2265 2.000e-07 ***
a.13        -17.24133    4.22811 -4.0778 4.812e-05 ***
a.14        -14.69461    4.23415 -3.4705 0.0005359 ***
a.15        -13.93474    4.23415 -3.2910 0.0010240 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares:    1143700
Residual Sum of Squares: 1032200
R-Squared:      0.097507
Adj. R-Squared: 0.086818
F-statistic: 9.12275 on 16 and 1351 DF, p-value: < 2.22e-16
> coeftest(a, vcov=vcovHC(a, type="sss")) ##This prints the regression results with cluster- and heteroscedasticity-robust standard errors

t test of coefficients:

             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  49.08410    3.42516  14.3305 < 2.2e-16 ***
Treatment    19.68941    2.81796   6.9871 4.393e-12 ***
a.1          -0.80603    0.28745  -2.8041  0.005119 **
a.2          -2.13215    0.53724  -3.9687 7.606e-05 ***
a.3          -3.89570    0.63600  -6.1254 1.184e-09 ***
a.4          -3.48976    0.68919  -5.0636 4.683e-07 ***
a.5          -4.06689    0.76424  -5.3215 1.204e-07 ***
a.6          -4.72398    0.85023  -5.5561 3.319e-08 ***
a.7          -5.27461    0.98815  -5.3379 1.102e-07 ***
a.8          -7.00221    1.05117  -6.6614 3.937e-11 ***
a.9          -7.22275    1.14087  -6.3309 3.309e-10 ***
a.10         -9.87291    1.19975  -8.2291 4.393e-16 ***
a.11        -35.05352    2.43158 -14.4160 < 2.2e-16 ***
a.12        -22.09806    1.64055 -13.4699 < 2.2e-16 ***
a.13        -17.24133    1.44338 -11.9451 < 2.2e-16 ***
a.14        -14.69461    1.51375  -9.7074 < 2.2e-16 ***
a.15        -13.93474    1.53586  -9.0729 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Functional Form for Pooled OLS with added controls:

$\text{bus\_journeys\_per\_capita}_{i,t} = \delta_1 + \delta_2 \text{d2011}_t + \delta_3 \text{d2012}_t + \ldots + \delta_{16} \text{d2025}_t + \beta\, \text{treatment}_{i,t} + \gamma_1\, \text{real\_med\_income\_pc}_{i,t} + \gamma_2\, \text{priv\_lic\_cars}_{i,t} + \gamma_3\, \text{tram\_jours\_percap}_{i,t} + u_{i,t}$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Results (with heteroscedasticity and cluster-robust SEs):

```
Pooling Model

Call:
plm(formula = Bus.Journeys.Per.Capita.Per.Year ~ Treatment +
    Private.cars.licensed.in.each.region.in.each.year + Journeys.on.light.rail.trams.per.year +
    Real.median.pay.in.each.region + a.1 + a.2 + a.3 + a.4 +
    a.5 + a.6 + a.7 + a.8 + a.9 + a.10 + a.11 + a.12 + a.13 +
    a.14 + a.15, data = BusJourneys_PrivateCars_TramJourneys_RealPay,
    model = "pooling", index = c("LA.or.Region", "Year"))

Unbalanced Panel: n = 87, T = 5-16, N = 1359

Residuals:
   Min. 1st Qu.  Median 3rd Qu.    Max.
 -52.64  -15.33   -4.62    6.23  139.35
```

```
Coefficients:
                                                  Estimate  Std. Error t-value  Pr(>|t|)    
(Intercept)                                      1.0021e+02  6.6491e+00 15.0716 < 2.2e-16 ***
Treatment                                       -5.5510e+01  1.8571e+01 -2.9890 0.0028495 ** 
Private.cars.licensed.in.each.region.in.each.year -1.4039e-02  3.1369e-03 -4.4756 8.267e-06 ***
Journeys.on.light.rail.trams.per.year            1.9876e+00  1.3877e-01 14.3234 < 2.2e-16 ***
Real.median.pay.in.each.region                  -2.4003e-03  2.9201e-04 -8.2201 4.752e-16 ***
a.1                                             -3.1439e+00  3.8468e+00 -0.8173 0.4139193    
a.2                                             -4.5941e+00  3.8372e+00 -1.1973 0.2314183    
a.3                                             -7.0076e+00  3.8442e+00 -1.8229 0.0685382 .  
a.4                                             -6.9396e+00  3.8488e+00 -1.8030 0.0716081 .  
a.5                                             -6.9374e+00  3.8513e+00 -1.8013 0.0718789 .  
a.6                                             -6.4488e+00  3.8436e+00 -1.6778 0.0936226 .  
a.7                                             -7.7834e+00  3.8351e+00 -2.0295 0.0426055 *  
a.8                                             -9.8477e+00  3.8382e+00 -2.5657 0.0104042 *  
a.9                                             -8.8533e+00  3.8424e+00 -2.3041 0.0213672 *  
a.10                                            -1.0128e+01  3.8257e+00 -2.6473 0.0082085 ** 
a.11                                            -3.4311e+01  3.8096e+00 -9.0065 < 2.2e-16 ***
a.12                                            -2.2905e+01  3.8480e+00 -5.9524 3.370e-09 ***
a.13                                            -1.9086e+01  3.8545e+00 -4.9516 8.298e-07 ***
a.14                                            -1.3954e+01  3.8098e+00 -3.6626 0.0002594 ***
a.15                                            -1.3222e+01  3.8097e+00 -3.4707 0.0005355 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    1137700
Residual Sum of Squares: 827450
R-Squared:      0.27273
Adj. R-Squared: 0.26241
F-statistic: 26.4279 on 19 and 1339 DF, p-value: < 2.22e-16
> coeftest(b, vcov=vcovHC(b, type="sss"))

t test of coefficients:

                                                  Estimate  Std. Error  t value  Pr(>|t|)    
(Intercept)                                      1.0021e+02  1.9209e+01   5.2170 2.105e-07 ***
Treatment                                       -5.5510e+01  2.1693e+01  -2.5589 0.0106108 *  
Private.cars.licensed.in.each.region.in.each.year -1.4039e-02  1.4213e-02  -0.9878 0.3234496    
Journeys.on.light.rail.trams.per.year            1.9876e+00  5.3608e-01   3.7077 0.0002177 ***
Real.median.pay.in.each.region                  -2.4003e-03  9.3912e-04  -2.5560 0.0106994 *  
a.1                                             -3.1439e+00  1.0541e+00  -2.9824 0.0029118 ** 
a.2                                             -4.5941e+00  1.2399e+00  -3.7053 0.0002198 ***
a.3                                             -7.0076e+00  1.5849e+00  -4.4213 1.060e-05 ***
a.4                                             -6.9396e+00  1.8066e+00  -3.8413 0.0001281 ***
a.5                                             -6.9374e+00  1.6378e+00  -4.2358 2.434e-05 ***
a.6                                             -6.4488e+00  1.5092e+00  -4.2731 2.064e-05 ***
a.7                                             -7.7834e+00  1.7852e+00  -4.3599 1.401e-05 ***
a.8                                             -9.8477e+00  1.9979e+00  -4.9291 9.294e-07 ***
a.9                                             -8.8533e+00  1.8560e+00  -4.7700 2.044e-06 ***
a.10                                            -1.0128e+01  1.5399e+00  -6.5771 6.859e-11 ***
a.11                                            -3.4311e+01  2.2868e+00 -15.0040 < 2.2e-16 ***
a.12                                            -2.2905e+01  2.0820e+00 -11.0015 < 2.2e-16 ***
a.13                                            -1.9086e+01  2.3043e+00  -8.2829 2.883e-16 ***
a.14                                            -1.3954e+01  1.5809e+00  -8.8264 < 2.2e-16 ***
a.15                                            -1.3222e+01  1.6122e+00  -8.2013 5.518e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Functional Form for Two-Way Fixed Effects transformation with controls:

Unobserved effects equation:

$\text{bus\_journeys\_per\_capita}_{i,t} = \alpha + \beta\,\text{treatment}_{i,t} + \gamma_1\,\text{real\_med\_income\_pc}_{i,t} + \gamma_2\,\text{priv\_lic\_cars}_{i,t} + \gamma_3\,\text{tram\_jours\_percap}_{i,t} + a_i + u_{i,t}$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Two-way fixed effects transformation:

$\ddot{\text{bus\_journeys\_per\_capita}}_{i,t} = \delta_1 + \delta_2 d2011_t + \delta_3 d2012_t + \dots + \delta_{16} d2025_t + \beta\,\ddot{\text{treatment}}_{i,t} + \gamma_1\,\ddot{\text{real\_med\_income\_pc}}_{i,t} + \gamma_2\,\ddot{\text{priv\_lic\_cars}}_{i,t} + \gamma_3\,\ddot{\text{tram\_jours\_percap}}_{i,t} + \ddot{u}_{i,t}$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

Results (with heteroscedasticity and cluster-robust SEs):

```
Twoways effects Within Model

Call:
plm(formula = Bus.Journeys.Per.Capita.Per.Year ~ Treatment + 
    Private.cars.licensed.in.each.region.in.each.year + Journeys.on.light.rail.trams.per.year + 
    Real.median.pay.in.each.region, data = BusJourneys_PrivateCars_TramJourneys_RealPay, 
    effect = "twoways", model = "within", index = c("LA.or.Region", 
        "Year"))

Unbalanced Panel: n = 87, T = 5-16, N = 1359

Residuals:
   Min. 1st Qu.  Median 3rd Qu.    Max. 
-73.489  -2.901  -0.334   2.941  25.000 

Coefficients:
                                                  Estimate  Std. Error t-value  Pr(>|t|)    
Treatment                                       -1.4410e+01  5.6106e+00 -2.5684   0.01033 *  
Private.cars.licensed.in.each.region.in.each.year -6.6603e-02  1.3947e-02 -4.7753 2.006e-06 ***
Journeys.on.light.rail.trams.per.year            9.2680e-01  1.2985e-01  7.1377 1.603e-12 ***
Real.median.pay.in.each.region                  -2.2221e-04  2.9323e-04 -0.7578   0.44870    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    61641
Residual Sum of Squares: 58184
R-Squared:      0.056078
Adj. R-Squared: -0.023022
F-statistic: 18.6099 on 4 and 1253 DF, p-value: 7.1698e-15
```

```
> coeftest(c, vcov=vcovHC(c, type="sss"))

t test of coefficients:

                                               Estimate  Std. Error t value Pr(>|t|)
Treatment                                   -1.4410e+01  7.3506e+00 -1.9604  0.05017 .
Private.cars.licensed.in.each.region.in.each.year -6.6603e-02  3.7101e-02 -1.7952  0.07287 .
Journeys.on.light.rail.trams.per.year        9.2680e-01  4.4285e-01  2.0928  0.03657 *
Real.median.pay.in.each.region              -2.2221e-04  5.4949e-04 -0.4044  0.68598
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Functional Form for First Differencing transformation with controls:

Unobserved effects equation:

$bus\_journeys\_per\_capita_{i,t} = \alpha + \beta\, treatment_{i,t} + \gamma_1\, real\_med\_income\_pc_{i,t} + \gamma_2\, priv\_lic\_cars_{i,t} + \gamma_3\, tram\_jours\_percap_{i,t} + a_i + u_{i,t}$

for i = 1, ..., 89, t = 1, ..., 16 (where applicable)

First differencing transformation (with aggregate time effects):

$\Delta bus\_journeys\_per\_capita_{i,t} = \delta_2 \Delta d2011_t + \delta_3 \Delta d2012_t + \ldots + \delta_{16} \Delta d2025_t + \beta\, \Delta treatment_{i,t} + \gamma_1\, \Delta real\_med\_income\_pc_{i,t} + \gamma_2\, \Delta priv\_lic\_cars_{i,t} + \gamma_3\, \Delta tram\_jours\_percap_{i,t} + a_i + \Delta u_{i,t}$

for i = 1, ..., 89, t = 2, ..., 16 (where applicable)

Results (with heteroscedasticity and cluster-robust SEs):

```
Oneway (individual) effect First-Difference Model

Call:
plm(formula = Bus.Journeys.Per.Capita.Per.Year ~ Treatment +
    Private.cars.licensed.in.each.region.in.each.year + Journeys.on.light.rail.trams.per.year +
    Real.median.pay.in.each.region + a.1 + a.2 + a.3 + a.4 +
    a.5 + a.6 + a.7 + a.8 + a.9 + a.10 + a.11 + a.12 + a.13 +
    a.14 + a.15, data = BusJourneys_PrivateCars_TramJourneys_RealPay,
    model = "fd", index = c("LA.or.Region", "Year"))

Unbalanced Panel: n = 87, T = 5-16, N = 1359
Observations used in estimation: 1272

Residuals:
    Min.  1st Qu.   Median  3rd Qu.     Max.
-95.6158  -1.4944  -0.0307   1.4999  42.7783

Coefficients:
                                              Estimate  Std. Error t-value  Pr(>|t|)
(Intercept)                                 4.2549e+00  1.8814e+00  2.2616 0.0238951 *
Treatment                                  -5.0987e+00  6.0238e+00 -0.8464 0.3974796
Private.cars.licensed.in.each.region.in.each.year -1.4440e-01  5.1907e-02 -2.7820 0.0054836 **
Journeys.on.light.rail.trams.per.year       1.1274e+00  1.1239e-01 10.0306 < 2.2e-16 ***
Real.median.pay.in.each.region             -3.1661e-04  2.4462e-04 -1.2943 0.1957952
a.1                                        -5.1218e+00  2.0213e+00 -2.5339 0.0114015 *
a.2                                        -1.0331e+01  3.8987e+00 -2.6499 0.0081534 **
a.3                                        -1.6212e+01  5.7801e+00 -2.8049 0.0051115 **
a.4                                        -1.9533e+01  7.6511e+00 -2.5529 0.0107997 *
a.5                                        -2.3800e+01  9.4890e+00 -2.5082 0.0122608 *
a.6                                        -2.7938e+01  1.1301e+01 -2.4722 0.0135592 *
a.7                                        -3.2169e+01  1.3155e+01 -2.4453 0.0146099 *
a.8                                        -3.7864e+01  1.5031e+01 -2.5190 0.0118923 *
a.9                                        -4.1643e+01  1.6868e+01 -2.4689 0.0136873 *
a.10                                       -4.7727e+01  1.8702e+01 -2.5519 0.0108308 *
a.11                                       -7.6205e+01  2.0592e+01 -3.7007 0.0002243 ***
a.12                                       -6.7733e+01  2.2392e+01 -3.0249 0.0025379 **
a.13                                       -6.7183e+01  2.4149e+01 -2.7820 0.0054831 **
a.14                                       -6.8076e+01  2.5914e+01 -2.6270 0.0087196 **
a.15                                       -7.1313e+01  2.7786e+01 -2.5665 0.0103889 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares:    119820
Residual Sum of Squares: 44088
R-Squared:      0.63207
Adj. R-Squared: 0.62648
F-statistic: 113.2 on 19 and 1252 DF, p-value: < 2.22e-16
```

```
> coeftest(d, vcov=vcovHC(d, type="sss"))

t test of coefficients:

                                                  Estimate  Std. Error t value  Pr(>|t|)
(Intercept)                                     4.2549e+00  1.5250e+00  2.7901 0.0053498 **
Treatment                                      -5.0987e+00  3.0133e+00 -1.6920 0.0908870 .
Private.cars.licensed.in.each.region.in.each.year -1.4440e-01  4.6889e-02 -3.0797 0.0021170 **
Journeys.on.light.rail.trams.per.year           1.1274e+00  4.9696e-01  2.2685 0.0234657 *
Real.median.pay.in.each.region                 -3.1661e-04  4.0286e-04 -0.7859 0.4320663
a.1                                            -5.1218e+00  1.7209e+00 -2.9763 0.0029735 **
a.2                                            -1.0331e+01  3.1894e+00 -3.2392 0.0012301 **
a.3                                            -1.6212e+01  4.7103e+00 -3.4419 0.0005966 ***
a.4                                            -1.9533e+01  6.1165e+00 -3.1934 0.0014410 **
a.5                                            -2.3800e+01  7.4411e+00 -3.1985 0.0014161 **
a.6                                            -2.7938e+01  8.8105e+00 -3.1710 0.0015559 **
a.7                                            -3.2169e+01  1.0213e+01 -3.1497 0.0016732 **
a.8                                            -3.7864e+01  1.1675e+01 -3.2431 0.0012136 **
a.9                                            -4.1643e+01  1.3008e+01 -3.2014 0.0014022 **
a.10                                           -4.7727e+01  1.4409e+01 -3.3123 0.0009519 ***
a.11                                           -7.6205e+01  1.6858e+01 -4.5203 6.758e-06 ***
a.12                                           -6.7733e+01  1.7868e+01 -3.7908 0.0001573 ***
a.13                                           -6.7183e+01  1.9196e+01 -3.4999 0.0004818 ***
a.14                                           -6.8076e+01  2.0455e+01 -3.3281 0.0008997 ***
a.15                                           -7.1313e+01  2.1901e+01 -3.2561 0.0011598 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Functional Form for One-Way Fixed Effects transformation with controls, time trend and covid dummy:

Unobserved effects equation:

$$bus\_journeys\_per\_capita_{i,t} = \alpha + \beta\, treatment_{i,t} + \gamma_1\, real\_med\_income\_pc_{i,t} + \gamma_2\, priv\_lic\_cars_{i,t} + \gamma_3\, tram\_jours\_percap_{i,t} + \psi_1\, t + \psi_2\, covid_t + a_i + u_{i,t}$$

for i = 1, ..., *89*, t = 1, ..., 16 (where applicable)

Fixed effects transformation:

$$\ddot{bus\_journeys\_per\_capita}_{i,t} = \beta\, \ddot{treatment}_{i,t} + \gamma_1 \ddot{real\_med\_income\_pc}_{i,t} + \gamma_2\, \ddot{priv\_lic\_cars}_{i,t} + \gamma_3\, \ddot{tram\_jours\_percap}_{i,t} + \psi_1\, \ddot{t} + \psi_2\, \ddot{covid}_t + \ddot{u}_{i,t}$$

for i = 1, ..., *89*, t = 1, ..., 16 (where applicable)

Results (with heteroscedasticity and cluster-robust SEs):

```
Oneway (individual) effect Within Model

Call:
plm(formula = Bus.Journeys.Per.Capita.Per.Year ~ Treatment +
    Private.cars.licensed.in.each.region.in.each.year + Journeys.on.light.rail.trams.per.year +
    Real.median.pay.in.each.region + t + Covid, data = BusJourneys_PrivateCars_TramJourneys_RealPay,
    model = "within", index = c("LA.or.Region", "Year"))

Unbalanced Panel: n = 87, T = 5-16, N = 1359

Residuals:
   Min.  1st Qu.  Median 3rd Qu.    Max.
-80.083  -2.815  -0.338   3.080  24.765

Coefficients:
                                                     Estimate  Std. Error  t-value  Pr(>|t|)
Treatment                                         -1.6379e+01  6.0076e+00  -2.7264   0.00649 **
Private.cars.licensed.in.each.region.in.each.year -6.1613e-02  1.4867e-02  -4.1442 3.637e-05 ***
Journeys.on.light.rail.trams.per.year              1.0716e+00  1.3934e-01   7.6904 2.931e-14 ***
Real.median.pay.in.each.region                    -2.0441e-04  2.6251e-04  -0.7787   0.43632
t                                                 -8.4041e-01  6.3243e-02 -13.2886 < 2.2e-16 ***
Covid                                             -1.6681e+01  6.4350e-01 -25.9231 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    170150
Residual Sum of Squares: 68351
R-Squared:      0.59828
Adj. R-Squared: 0.56909
F-statistic: 314.243 on 6 and 1266 DF, p-value: < 2.22e-16
> coeftest(e, vcov=vcovHC(e, type="sss"))

t test of coefficients:

                                                     Estimate  Std. Error  t value  Pr(>|t|)
Treatment                                         -1.6379e+01  7.7302e+00  -2.1189   0.03429 *
Private.cars.licensed.in.each.region.in.each.year -6.1613e-02  3.7850e-02  -1.6278   0.10382
Journeys.on.light.rail.trams.per.year              1.0716e+00  4.5844e-01   2.3375   0.01957 *
Real.median.pay.in.each.region                    -2.0441e-04  3.6465e-04  -0.5606   0.57520
t                                                 -8.4041e-01  1.4624e-01  -5.7467 1.139e-08 ***
Covid                                             -1.6681e+01  1.3270e+00 -12.5703 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Section 5: Limitations

As noted in Section 3 of the main text, I would ideally like to control for per capita income, household car ownership, pricing of competing and complementary public transport services, demographic makeup of the area and population density of the area – there are likely further factors that I would like to include in my regression too. I haven't been able to control for even these named factors fully: there is currently missing data on demographic makeup and population density of the local authorities, potentially leading to omitted variable bias in models (2) to (5), and imperfect proxy variables have been used in order to control for household car ownership and pricing of competing and complementary public transport services. If we are unwilling to assume that any difference between the proxy variable and the unobserved variable of interest is uncorrelated with the value of the proxy variable, this could lead to further bias in the estimated treatment effect in models (2) to (5) – compared to pooled OLS, fixed effects and first differencing models are particularly sensitive to this type of bias as the difference between the proxy variable and the true value of the control can be large compared to the time-demeaned or first differenced control variable (Wooldridge, 2025, pp. 469, 487). Moreover, most of the datasets had small amounts of missing data; if the reason for missing data is correlated with the value of the dependent variable, this is analogous to having a non-random sample and will lead to further bias in the estimate of the treatment effect (Wooldridge, 2025, pp. 328–331).

Another potential problem is that there are only 88 cross-sectional units included in the regression. Depending on the distribution of the variables in the error term in the population, this may not be a large enough cross-sectional sample to use asymptotic justifications for test statistics and confidence intervals (Wooldridge, 2025, ch. 5). While this doesn't affect the consistency of the results, it calls into question inference conducted throughout the paper, as noted in section 3 of this Appendix.

Section 6: Conclusion

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